Candidate seat no.:_____

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B.Tech. Theory Examination

April - May 2018

MA142 Engineering Mathematics II

Date: 30.04.2018, Monday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions: 1) The question paper comprises two sections. 2) Section I and II must be attempted in separate answer sheets. 3) Make suitable assumptions and draw neat figures wherever required. 4) Use of scientific calculator is allowed. 5) Candidate must write exam seat number on paper.

SECTION - I

Q-1 **Fill in the blank by choosing the most appropriate answer from the given options.**

(a) If $(2x+ay)dx+(5x-6y)dy=0$ is an exact differential equation, then the value of a is ______. **01**

(i) 2 **(ii)** 3
(iii) 4 **(iv)** 5

(b) The degree of differential equation $\sqrt{y'''}-5y''+6y'-y=0$ is ______. **01**

(i) 1 **(ii)** 2
(iii) 3 **(iv)** 4

(c) The Bernoulli differential equation $y'+p(x)y=q(x)y^n$ reduces to a linear differential equation, if n is ______. **01**

(i) 1 **(ii)** 2
(iii) 3 **(iv)** 4

(d) The general solution of a differential equation $y'=2x$ represents family of ______. **01**

(i) straight lines **(ii)** circles
(iii) parabolas **(iv)** hyperbola

(e) If general solution of a higher order homogenous ordinary differential equation contains four arbitrary constant, then order of that ordinary differential equation is ______. **01**

(i) 1 **(ii)** 2
(iii) 3 **(iv)** 4

(f) One of the solutions of nonlinear partial differential equation $p+\sqrt{q}=1$ is ______. **01**

(i) $z=(1-b)x+by+c$ **(ii)** $z=(1-\sqrt{b})x+by+c$
(iii) $z=(1-b)x^2+by^2+c$ **(iv)** $z=(1-\sqrt{b})x^2+by^2+c$

(g) One of the solutions of nonlinear partial differential equation $z = px + qy + (pq)^{3/2}$ is ______. **01**

(i) $z = ax^2 + by^2 + (ab)^3$ **(ii)** $z = ax + by + (ab)^3$

(iii) $z = ax + by + \sqrt{ab}$ **(iv)** $z = ax + by + (ab)^{3/2}$

Q-2 **Attempt any *four.***

(a) Find the general solution of $(xy^3 + y)dx + (2x^2y^2 + 2x + 2y^4)dy = 0$. **04**

(b) Obtain the general solution of $y'' + 2y' + y = 3e^x$ using variation of parameter method. **04**

(c) Solve $y'' + 5y' + 6y = e^{-3x} + x$ using method of undetermined coefficient. **04**

(d) Form a Lagrange's linear partial differential equation satisfied by $f\left(x^2 + y^2,\, z - xy\right) = 0$. **04**

(e) Solve $x^2 p + y^2 q = (x + y)z$. **04**

(f) Define partial differential equation and construct a nonlinear partial differential equation by eliminating arbitrary functions f and g in $z = f(x + at) + g(x - at)$. **04**

Q-3 (a) **Attempt any *two.***

(i) Find the particular solution of $y' + \dfrac{y}{x} = 2x^3$ using initial condition $y(0) = 1$. **03**

(ii) Find the general solution of $x^2 y'' + xy' + y = 0$. **03**

(iii) Identify type of nonlinear partial differential equation $p^2 + q^2 = x + y$ and solve it. **03**

Q-3(b) **Do as directed**

(i) Show that $(y^2 e^{xy^2} + 4x^3)dx + (2xye^{xy^2} - 3y^2)dy = 0$ is an exact differential equation and derive its general solution. **03**

(ii) Construct a second order ordinary differential equation satisfies by $y = Ae^x + Be^{2x}$, where A and B are arbitrary constants. **03**

OR

Q-3(b) **Do as directed**

(i) Find the general solution of $y''' - y'' + 100y' - 100y = 0$. **03**

(ii) Identify type of nonlinear partial differential equation $z = p^2 + q^2$ and solve it. **03**

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SECTION – II

Q-4 **Fill in the blank by choosing the most appropriate answer from the given options.**

(a) If T be the temperature of the body at time t and t_s is the temperature of surrounding, then Newton's law of cooling is given by ____. where k is constant. **01**

(i) $\frac{dT}{dt} = k(T - t_s)$ **(ii)** $\frac{dT}{dt} = kT + t_s$

(iii) $\frac{dT}{dt} = kT$ **(iv)** $\frac{dT}{dt} = k(t_s - T)$

(b) A function $u(x, y) = e^x \cos y$ satisfies a partial differential equation of the form ______. **01**

(i) $u_x + u_y = 0$ **(ii)** $u_{xx} + u_y = 0$

(iii) $u_x + u_{yy} = 1$ **(iv)** $u_{xx} + u_{yy} = 0$

(c) $\int_{-1}^{1}\int_{-2}^{4}\int_{2}^{3} xy^2 z^3 dxdydz$ = ______. **01**

(i) -125 **(ii)** 0

(iii) 136 **(iv)** 155.656

(d) $\iint_R r dr d\theta$ = ______, where R is the region bounded by curves $x = 0$, $x = 1$, $y = 0$ and $y = 2$. **01**

(i) 1 **(ii)** 2

(iii) 3 **(iv)** 4

(e) The sum of error function and complimentary error function is ______. **01**

(i) 0 **(ii)** 1

(iii) 2 **(iv)** 3

(f) If $P_n(x)$ is a Legendre polynomials for $n \in \mathbb{N}$, then simplified form of $\frac{2}{5}P_3(x) + \frac{3}{5}P_1(x)$ is ______. **01**

(i) x **(ii)** x^2

(iii) x^3 **(iv)** x^4

(g) The value of $erf(x)$ at $x = 0$ is ______. **01**

(i) 0 **(ii)** 1

(iii) 2 **(iv)** 3

Q-5 **Attempt any *four.***

(a) Find the possible solutions of $u_t = c^2 u_{xx}$ using method of Separation of variables. **04**

(b) A constant electromotive force E volt is applied to a circuit containing a resistance R and an inductance L. If initial current is zero, then show that the current builds up to half of its theoretical maximum in $LR^{-1}\log_e 2$ seconds. **04**

(c) Evaluate $\int_0^\infty \int_0^\infty e^{-x^2-y^2}\, dxdy$ by changing it into polar coordinates. **04**

(d) Evaluate $\iiint_V 2x\, dzdydx$, where the solid region V is bounded by plane $x=0,\ y=0\ \ z=0$ and $x+y+z=1$. **04**

(e) Show that $J_{\frac{1}{2}}(x)=\sqrt{\frac{2}{\pi x}}\sin x$, where $J_n(x)$ is a Bessel function of order n. **04**

(f) Express $f(x)=x^4-2x^3+3x^2-4x+5$ in terms of the Legendre polynomials. **04**

Q-6 (a) **Attempt any *two.***

(i) The differential equation of a simple pendulum is $\frac{d^2x}{dt^2}+\omega^2 x=e^t$. Find motion($x$) in terms of t. **03**

(ii) Evaluate volume of a cylinder bounded by $x^2+y^2=4$, $z=1$ and $z=2$ by using concept of triple integration. **03**

(iii) Show that $\int_0^n x^n(n-x)^p\, dx=\frac{n^{n+p+1}\Gamma(n+1)\Gamma(p+1)}{\Gamma(n+p+2)}$ for $n>0$ and $p>0$. **03**

Q-6(b) **Do as directed.**

(i) Evaluate area of a triangle having vertices (0, 0), (1, 1) and (0, 1) by using the concept of double integration in Cartesian coordinates. **03**

(ii) If $\{P_n(x);\, n\in\mathbb{N}\}$ is a set of Legendre polynomials, then show that $P_n(1)=1$ and $P_n(-x)=(-1)^n P_n(x)$. **03**

OR

Q-6(b) **Do as directed.**

(i) Find the charge q at any time t from the differential equation $L\frac{d^2q}{dt^2}+\frac{q}{C}=0$ of LC-circuit. **03**

(ii) Show that $erf(x)=\alpha(x\sqrt{2})$; where $\alpha(x)=\sqrt{\frac{2}{\pi}}\int_0^x e^{\frac{-t^2}{2}}\, dt$. **03**

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